

ActInf Livestream #015.1 ~ “Free-Energy Principle, Computationalism and Realism: a Tragedy”

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[Music] hello everyone welcome to the active inference lab to the active inference live stream today is active inference live stream 15.1 and it is february 2nd 2021 welcome to the active inference lab everyone we are an experiment in online team communication learning and practice related to active inference you can find us on our website twitter discord gmail youtube or keybase public team this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and as far as video etiquette for live streams we'll mute if there's noise in our background and raise our hands so we can hear from everybody today we are in 15.1 on february 2nd and really excited and thankful to have both authors of this paper today and also joining us for next week so that's going to be really exciting check out this spreadsheet if you want to see the most updated event calendar for active inference live stream but here we are in 15 with a rapid introduction today we're gonna have a introduction to each other especially some of us who are new on the stream and then we're going to be kind of walking through the paper having ideas having fun conversations and also writing down ideas that we want to discuss more in 15.2 which will be like a follow-up discussion on this so if you're participating live or watching later then just write down ideas and questions you're having and get in touch with us if you want to participate so here we are in the introductions today we'll go around introduce ourselves and then just we're in it so we'll have a fun conversation um i'm daniel friedman i'm a postdoctoral researcher in california and yeah i'm just really excited to be having this conversation so i will pass it first to our authors ines and thomas up we can't hear you in s but you're not muted so reload and let's go to thomas thomas can you unmute and introduce yourself and then and that's just like re like control r reload and then you'll probably work with the audio but go ahead thomas yeah um hi everyone my name is thomas from s um i'm currently a phd student in antwerp at the university of antwerp um i'm working primarily on inactivism and free energy principle thanks for having me thank you

we're gonna look forward to hearing a lot more from you and s can we hear you now okay can you hear me now perfect great all right um all right so my name is ines um i am a postdoctoral fellow at the berlin school of mind and brain and yeah and my my topic of research is on dynamical systems and e-cognition in in general great um then we can just pass it to whoever hasn't spoken so go to shannon hey i'm shannon i'm a phd student at the university of california in merced um and i work on sensory motor neuroscience and the dynamics of crowd behavior um i'll pass it to sasha i'm here let's go to steven then scott david oh hello i'm stephen i'm in toronto i'm i work a lot with participatory theater and community development i'm doing a practice-based phd at the moment through canterbury christ church university and i will pass it over to alex hi everyone i'm alex i'm in moscow russia and i'm a researcher and systems management school and still trying to find ways to join active influence and systems thinking frameworks and i pass it to session hi um i'm sasha i'll try this again i'm a graduate student in neuroscience and studying early brain development and based in california and i will pass it to blue good morning i am blue knight i am an independent research consultant based out of new mexico i don't know who hasn't gone yet maybe i'll pass it to marco hi uh i'm marco marcos i'm based in holland um basically just a free energy active influence enthusiast um happy to be here i'll pass it to scott hi folks um my name is scott david i'm the director of the information risk research initiative at the university of washington applied physics lab my background is a lawyer and what we're doing is engineering systems that are socio-technical systems so it's parameterization for technical and social elements of active inference based systems thanks well thanks everyone so much for coming on live this is really going to be a great discussion and also a really prescient paper so great to hear everyone's views on it in these warm-up questions we can just begin where wherever people have been heading heading into today's discussion so i'll just list them and then people can raise their hand and we'll hear from everybody who wants to speak what is something you're excited about today what is something that you liked or remembered about the paper what motivated you to come to this discussion to read the paper think through these topics and then also what's something that you're wondering about or would like to have resolved by the end of today's or next week's discussion so until anyone raises their hand um ines and or thomas um i just wanted to ask how do you start by communicating this research or what led you to this question i'm going to kick out the ghost in esses but um um yep so she should i take this or do you wanna but you're muted right now sorry um yeah no thomas just go ahead yeah yeah um so i think the way i got to the the idea of the paper is that i'm um primarily coming at it from a radical inactivist

perspective um where there's a lot of um antagonism towards representations models and whatnot um in the meantime i got very excited about free energy principle and i've been sort of trying to sort out where that all fits what the models mean what everything means in everything that's being said um and so ines and i were talking about it and um we started talking about representations and models in the free energy principle and we figured out that it really depends on how you interpret the models and what exactly you take the mathematics to mean um and based on that we just started listing out the different options you could have and um what they mean and what we personally thought about those what we thought was were good aspects less nice aspects um and through all of that we sort of came up with instrumentalism as the uh only path forward awesome i'm sure we're gonna have a lot of great second thoughts and conversations on that so anyone can raise their hand but either an s or thomas just to lead into those who might not be familiar with long-running debates in the philosophy of science or multiple possible ways to interpret formalisms how do you convey this instrumentalism realism question and motivate its importance shall i take it thomas sure okay um so uh one thing that we thought was like really interesting um to think about is that um the majority of the the accounts within the free energy principle or process theories uh they are very much realist so that's what we thomas and i started questioning and we think that it's it's interesting because uh there's been other people also joining that particular consensus that we also have um so we thought that this would be something that we needed to look at uh and the question just to sort of like give a little bit of context on where we are in the literature is that the majority of the approach the approaches were um on the realist camp even those that were um coming from an activism and that's why we found things to be a little bit strange because the first question is whether one is licensed to take a realist account from the the fact that we can model uh the brain or we can model the neural activity or we can model um uh cognitive cognitive activity so that's the one question that we wanted to to think about and look at whether we are licensed to take a realist stand and if we are what is the argument what is the evidence to think that um this machinery that we we use in our scientific model modeling um can also be uh exists somewhere in uh neuronal activity or cognitive activity so that was one thing then the other thing was that it start there started to be like a sort of like a wave that was taking um the free energy to be compatible with an activism and um and it it might we started thinking about and questioning this because it might be the case that they may not be compatible after all because as we know on the side of the in activism uh there's like a claim um against uh representations for example then on the predictive uh accounts of

cognition on the other side you sort of need representations to get your theory going because you need to have predictions you need to have prediction errors coming so you need to rely on representational machinery so that's what we thought that it's important for us to question and think about whether it is actually the case that um these two things can be made compatible so on the one side the predictive accounts of cognition which take uh representations in a very literal account um and on the other side the inactivism uh which uh for the majority of the people on an activism side uh representation cognition doesn't come down or is not all the way through representational so this is sort of like where we are navigating the literature here it's sort of like between the very representational um predictive accounts of cognition and um and the realism that that's a stand that they kind of need to take and on the other side the inactivism which there's sort of like a tendency uh recently to be made like sort of compatible under the same roof so to speak great thank you both for contextualizing and everybody can raise their hands so that we can make sure to hear from everybody's excitement here um but yes the bigger question is does making a model have a realist interpretation if you make a model of the brain as a steam engine is it really a steam engine no but then if you make a model of it as a free energy principle corollary process theory is it really that that's kind of the discussion and there's a lot on the line which we're gonna tease out so shannon and then anyone else who raises their hand yeah so sort of jumping to the end of the paper already you say like the model in scientific practice is a representation of the nervous system to the extent that it holds explanatory capacity and i think um like a lot of neuroscientists that i talked to if you start bringing up you know the representation or the representation debate that's like ah that's something that philosophers worry about um but we know when we say representation we just kind of need a word stand in for our conversation so we can talk about um like patterned activity of neurons and we don't care that it could stand separate or not the situation or the environment but it's useful it's a useful word to use so that way we can have language and talk about the system or the pattern of activity that we're interested in that's associated with some behavior um and i i just think that sounds right like how you conclude it and maybe i didn't have a question maybe this was a moment where i had to comment more of a question so apologies but if that sparks some conversation that would be great it's something that stayed with you and will change how you think about even other kinds of models next time you see a linear aggression maybe you're going to wonder if it's being used in a realist or an instrumentalist context so thanks for sharing it shannon stephen and then anyone else sort of one thing i think is quite interesting is the just like shannon was saying about you know often a

word gets used but then it's kind of fuzzy and then it kind of gets like hand waved away around why it's maybe not that clear i think this also may be the same challenge comes up with pragmatism because pragmatism is really useful and it gets used in a lot of context but now because everything's slightly pragmatic in the sense that it's all about action through the free energy principle it becomes so cool instrumentalism sort of helps in a way get a bit of clarity when it's something which is about theory so i wonder i'd be interested to know about how the pragmatism um you know wasn't used as the word and used a more sort of clear unpacking i suppose yeah great question so anybody authors otherwise what is the relationship between pragmatism and utility and then instrumentalism so we're using the instruments ostensibly for pragmatic ends but the authors chose to frame the discussion in terms of realism versus instrumentalism instead of pragmatism versus say anti-pragmatism or some other you know could be pragmatism versus realism perhaps so any authors or other hands um i i don't mind saying just a few words um and then perhaps i'll end it over to thomas but i just want to um the one way that i think that it's useful for me to think this is um the use of representation so in what sense are we using representations because um shannon was saying she was raising a very good point it is very useful to refer to these things that this activity that's going on in in the brain as a representations but i think that um what in what sense are using representations is important because if it's in a sort of a metaphor then obviously there is no problem because it is actually very useful and we are building all of these we are trying to make sense of all of these data that we gather from the brain uh and that's in some in some sense building a representation because what is a representation representation is a is a sentence it's a picture it's a map these are kinds of representations why is this problematic it's because these are things that we as human beings do we do that in our daily lives and we do that in our labs so we build these representations to make sense of uh the observations that uh that we that we we do in our labs and in our daily lives so then we can obviously use this word this concept but it doesn't have the same meaning or it does in the realist sense so i think it comes down to how serious are we in the use of these of these are we using this in a literal sense in the sense of like sentences maps pictures that we as human beings have and build and develop and the brain also has them or are we using in a metaphorical way which is if it's in a metaphorical way because it's useful to use the concept then it should be fine it's a metaphor okay and that's it really interesting a metaphor is really an instrument when somebody says that this movie was you know was like it's like an ocean of ideas or it was like an ocean a simile they're using that sentence representation instrumentally to convey a point to you know communicate

something that is internal to the thinker and then that is not meant to be interpreted realistically but in science where there's a strong kick amongst some people they think it isn't science unless it's not gonna be the realest most mechanistic thing now whether it's mechanism all the way down or what what the lowest level mechanism is that's sort of where it the those realist ideas bottom out but without going that way it's just it is across fields and across different media we ask whether we're meaning something really or instrumentally so any other thoughts what was some like uh we have oh yeah scott people can raise their hand on shitsy because i'm not always looking at the actual video screen so thanks scott and then anyone else i was just wondering it's it's interesting um in terms of the encoding and the decoding notion so if you have different isms on either side of your encoding and decoding that seems like where you have the problem potentially so someone's encoding a communication as a metaphor but it's taken not as a metaphor but as a something stronger and is decoded that way then then there's a miscommunication one of the things that i've been trying to fuss with and it feels like this is that place the painter kandinsky said that violent societies yield abstract art and i always wondered whether that's reversible whether abstraction is a form of violence and here that feels like that might be relevant because if i abstract something with one ism and then it's in my encoding of my message and then it's decoded by someone else who's decoding with a different ism then that abstraction has led to a violence in fact it might lead to actual harm to that other person because they may misinterpret my message based on that schism of isms that you have there i had to go there but um anyways is that something that that challenge of communication across boundaries feels like it's invoked here and what might be some strategies for doing something about that thanks scott for the great question thomas and then we just continue down the stack so people can just throw up the hand raise whenever they have a thought and we'll get to everybody yeah i think um these these three comments really uh together cut at the heart of the matter um i think that indeed from what i have heard myself neuroscientists pretty much often know what they mean with and they don't take it to mean anything much more robust than a sort of correlation between particular patterns of activity and particular things that we see happening um and it's more robust than that but not anyway what the philosophers take it to mean um but the issue is then exactly in communication where um it feels like among neuroscientists it's a bit of an in insiders knowledge that representation is sort of a metaphor it's like the computer metaphor as a whole um it's a metaphor and you shouldn't take it uh in a realist way as we would say um but when students read a handbook this is not always that obvious and i've also talked to neuroscientists that really believed that

the rats had a literal map in their brain like we have a sort of jpeg on our computer um because they were talking about representational maps in the brain and if you know a little bit about philosophy in this area philosophers take these new scientists talk about representations to mean just about anything that they like um so i think that the the issue is indeed one if you want to talk about that way of encoding and decoding that they encode a metaphor and some people that are in the know decode a metaphor but many people don't and i've noticed that exactly with the free where it feels like it gets very muddy with what is metaphor and what isn't and what people take to be metaphor and what isn't um and that sense my my main project is is to try and create clarity um in what it could possibly mean and even if people completely disagree with the position i think at the very least i would like just people to be more clear about what they really mean thanks thomas and really one of the goals of these discussions that we're having so we're going to go to shannon stephen and s yeah this is actually really great um thomas and scott i hadn't thought about it this way before so no one in the general public cares about the representation words that might not be true but like nobody cares but whenever science from a neuroscientist gets picked up by a journalist who then makes the article with the headline that says your brain has models inside of it that it's running and you should learn music because it will make you smarter but learning coding won't make you smarter which was like a headline recently um but that translation process that scott was talking about that encoding and really gets stuck and it doesn't get decoded when it gets written about in sort of like more mainstream news outlets like maybe someone's medium blog post that they're writing and they're a really dedicated grad student who's grappling with these ideas all the time and they're very clear in their writing about what scientists mean when they're talking about a representation or a model what this means for the actual brain itself and then what this means for you as a person just reading about it um but that doesn't always come across in like the headline on cnn or or like the the the facebook feed that's scrolling through you know my adult parents computer screen and so being clear you have to go through this step of using it as a useful metaphor in neuroscience getting clear what the metaphors could be if they're real or just instrumental readings in the philosophy camp and then like the one more step is bringing the science communicators in and and getting them to to parse this huge debate down into something that's able to be distributed to folks who aren't enmeshed in this every day thanks shannon so stephen and that's sasha and well this clash is definitely um a useful discussion because i think in activism radical inactivism and free energy are both trying to come in and say it's not all in the head so just like you're saying this the challenge you've

got is anyone and as soon as people talk about neuroscience and that it's like there's just an assumption in our culture that's all in the head it's not even so that metaphor is like a signpost and then everything else becomes and is it is it a film playing in my head and it's like well wait a second it's the mind-body environment with the dynamic system it's about action and they look at you with this like wait so it's like it's like it slips into that so i think this is a really good good point because like metaphors when you embody them are kind of like you inactively understand what it's like to be inside them but then when they just become like a signpost for what to do they lose all of that kind of architecture so i think this is this is quite valuable yep the metaphors and the language stephen and related to the science communication point it's uh oh well this is hardwired in or the dna codes for this protein it uses the language that's implicit in a very specific understanding and that metaphor gets over interpreted and then you start where's the ram in the brain you know where's the gpu and then you start building out this metaphor before you know it you're way down the rabbit hole with something that was basically in terms of this paper taking a realist reading on an project and again we just think about other kinds of modeling if you did a steam engine model of the brain it's not a steam engine if you do a linear regression it's not a linear regression so if you do an active inference model it's not an active inference brain but that is very nuanced to communicate because of the way that our language is so ines and sasha than anyone else yeah um i think these are very good remarks i um just wanted to um i think i have to disagree just a tiny bit just to make a point about i i think that there are real issues and there are real dangers in conceiving of uh in the ways that we conceive things and especially in the ways that we use our language and our concepts and particularly looking at the concept of representation but there are others that i can think of i can think of the concept of hierarchy for example that could be a metaphor could be used in a realist um way uh there there's a there's new papers on this which are very relevant and very interesting but what i want to say is that in the particular case of the representation i think there are real dangers in using it's not just how the science is communicated or oh maybe this is not a problem for me to deal with let's just philosophers handle that that's not real for my particular lab it's not going to be an issue i think it's going to be an issue because i think the way that you conceptualize however the processes are going on in the brain is going to dictate the way that you design and you interpret the results so i'm thinking for example there's a bit in the paper where we do we make a distinction between two kinds of modeling techniques that we can use and they have two very different goals ambitions worries and they end up in very different conclusions and um so one of them um takes up as

for free that there is information in the brain so this kind of modeling is the kind of modeling that uses structural causal models and uh what they aim for is to think about uh the the relations between the nodes uh these are nodes that are highly connected and the way that we design and understand or interpret the data coming from those models is by the presence or absence of information now this means that we are taking information for free information for existing in the brain so what kind of information are we talking about is it the kind of information that i also have in my model because if it is then we're talking about semantic information and i mean it may be the case that the brain actually works with the very same kind of information that i have in my model but we need to prove that that's an assumption and this is how we are building our models and i think that's that's one worry now the other worry is that in building these models that that's the inactivist worry uh which stephen was just mentioning which is if we say that well the cognition comes down to this representation representations happening in the brain then so much for the rest of uh cognition that is uh in the body in the body being in an environment so that this old is an activist sort of like tradition and i just wanted to add that cool thank you um we have sasha shannon alex um yeah thank you for explaining that so well um and bringing this to the forefront that this is an issue to be grappled with i think um from the neuroscience background uh yes it is just relegated to uh philosophers to deal with uh these bigger questions but um leading to this uh going back to this idea of encoding and decoding it really does matter what is meant by the phrase and then how it's um decoded and i think in the world of like molecular neuroscience and um just recording signals from the brain we really think that um we're measuring something real about the system whether or not that's what's actually happening that's just one measurement of the overall behavior and um to just lean into instrumentalism it's a useful metaphor for what we're seeing but um obviously it people don't have the training or the decoding of the philosophy perspective on what that means and so um i've never thought about it until this paper um that this could be a real issue even in my own work so i think it's really important to um get into the nuance of what we mean when we use these metaphors thanks sasha shannon alex well okay yeah go ahead shane um yeah so that's really great i i didn't mean to say that like only science communicators and philosophers are should worry about this question um but i think so like something that that this reminded me of in sort of my field so i look at music perception um in the brain and i kind of have a question of does the brain use internal models to predict timing mechanisms and this metaphor that you brought up that how it affects the assumptions that you make and how you design or analyze or interpret your experiment um i

think it also makes a case for pluralism like if you're realist then pluralist about realist approaches if you're making metamorph metaphors then pluralist about the type of metaphor that could be useful so we thought or a lot of neuroscientists still think there's a clock mechanism in the brain or a pacemaker that's just like tracking rhythmic activity but maybe there's this move now that there's more of a bayesian mechanism that's building up a timing model that's now predicting when or what the next event will be that occurs and one of those might be more explanatory useful and the other one will fall away but i don't know if we can actually ever prove that there is a bayesian mechanism in the brain or that there is a clock model in the brain i think as far as we can get is that taking this approach is more explanatorially useful than taking the other approach cool and it made me think from my as an evolutionary biologist about the question what is a gene and that's a philosophical question that they don't really discuss too much in genetics departments but of course the way that you consider what a gene is it's going to set up what you measure who's included on the research project how you write about it how you analyze it i mean it's the primary determinant so whether what's happening with the electrical signals in is part of a real model that is being built out or whether that is something like we're just measuring the temperature but then we're modeling something totally separate and then also just quickly um the pluralism it's easier to be pluralistic with metaphors because it's like we can always write new poems we can always have new metaphors we can always build and do yes and with the metaphors but realism it's a little harder to be pluralist because ostensibly it's not multiple things at once or potentially you know we can't all be living our own truth for what the brain is but we of course can have our own perspectives so actually it totally opens the door to pluralism in a really qualified way where we can come together to talk about what's useful rather than debate about who's right about how things are so really nice framing so alex and then in s thanks for me as i came from engineering world to learning active inference instrumentalism approach was an only way starting to understand what what is happening uh because uh when i started to for finding and look at uh models uh in this framework it was a way and to for my mind to get the basics and trying to go deeper uh but as for this paper for me it's great because it gives great promotion for instrumentalism approach and i just think if there is a notion for radical instrumentalism if so can we use it and how it should be described uh and again for me honestly i don't care much about philosophy debates or even neuroscience debates i'm watching on a world around in terms of systems and trying to understand these systems basically as black boxes understand their behavior without understanding how this deep rabbit holes inside working so that's why

it's interesting for me is for reading this paper and with discussions too thanks thanks alex so
ines and then thomas yeah um i just wanted to add something to what you were just
mentioning danielle um about um the the pluralist approach um i think that yeah you were you
said that something that sounded very very right to me which is one can only take this pluralist
or is it at least easier to take the pluralist route um if we are on the sort of the side um so for
example uh with the example that we have we have in the paper with these two kinds of
different modeling approaches that we can take like dynamical causal modelling or structural
causal modeling if we take an instrumentalist approach then we can say that these two
different sorts of modeling techniques they can work together where for example what you get
from the structural cause of modeling which is you get like a sort of a nice topological
description of the connections arranged in the brain as we get a nice map of that which can
sort of combined with dynamical causal modeling where you get for example the dynamics in
terms of states and how the activity occurs you can combine this in a pluralist way as long as
you take this much more instrumentalist approach and you don't think you know it to be the
case that what you're getting from these models is what is actually the case so i just wanted to
add that yes like we have one instrument that helps us find the structure and topology the
connectness we have another instrument that measures or helps us do inference on a slightly
different attribute and it's not a philosophical debate whether it's actually just dynamics or
actually just topology it's actually us investigating a system we care about and then that's what
can bring people to the table so thomas and then scott david than anyone else yeah um i think
that's i think that's really quite right um i've spoken about this paper with other people as well
and some were worried um that instrumentalism was placing too strong of a limit on our
scientific endeavors that if we say oh well we could only ever be instrumentalist then they
would they said why do science at all if we can't really get at the heart of the matter if we can't
actually add some sort of finish line say okay well now we have just an accurate description of
what cognition is and how organisms navigate the environment um and i think that's a fair
worry um i wonder if perhaps it's just a bit too ambitious uh but you know who knows what
we can do in 100 years and i think i do think in that perspective it is important to see that the
paper doesn't really get into that grander philosophy of science debate about model
instrumentalism across the board right so it could very well be that at some point we'll get a
model that really well describes everything that really happens in in an organism maybe the
main point here as i take it anyway might disagree of course is that at the very least in the free
energy principle you don't get that entailment of because we can model it the organism models

as well um but it's not it could be that in the end we can take a realistic position on some other model in the future nice so we're not committed to yet this instrumentalism across the board we can be realist elsewhere and we can be realist in the future but you've laid out a research agenda and a logical structure for asking whether you know have we met some of these claims that were brought up in your paper on another topic or for free energy in the future on that note to quickly get back to what alex was saying it doesn't necessarily mean that we have to commit to radical instrumentalism we don't have to be that radical although that might be called engineering by some so maybe it already exists so scott and then in s yeah and along those lines so i mentioned before that i was an attorney rhetoric is persuasive speech and it is what we attorneys produce and so it's interesting when you talk about instrumentalism and realism in terms of rhetoric so i used to say to my kids all the time that the only time you see reality is when you see paradox and any other time you're living inside a model and so that means that when people talk about resolving paradox what they're talking about is whose model wins but i always told my kids you should manage paradox not try to resolve it because that is reality so let me take that what i'm suggesting is metaphor may be reality in certain contexts because the management of the paradox happens through rhetorical speech the incommensurables are brought together with with with the rhetoric you you create a story to make it a thing right and so what i'm wondering is i'm i was thinking of the analogy of the well-tempered clavier when bach was tempering musical notes right there's all sorts of notes out there and then the scale was decided upon and now it's a scale and it's objective and it's mathematically demonstrable why it's a good scale but it's not the only scale and so that kind of idea of managing paradox through i guess it's instrumentalism because you're allowing for multiple approaches what i'm wondering is is there a preference for one or the other um in in trying to motivate change in in in groups and then second part of that so that's one and then the other part is i beginning to understand that the i believe that the mind doesn't exist in the brain it exists in language and the brain is an antenna that's tuned into that so from that perspective can anything be other than instrumentalist so you made a claim about what the brain is so that was a realist claim about it but i yes great questions scott so we got an s blue um okay i just wanted to um add something to uh what thomas was saying actually uh whether we lose something when we take the instrumentalist part i think that my concern is that i'm not sure if we can if we can um if you can even have the realist path and i don't i'm not sure if that's an option um so the reason being and i'm going to come back to the examples that we have in the paper because i think these are like practical you know useful examples are concrete examples

that we can think about amongst ourselves even if we come from very different backgrounds but uh there are some scientists here some neuroscientists here and i think that's useful to you know come back to uh what we're really talking about uh with examples so with the two different kinds of modeling or let's say just two major ones that we refer to in the paper where uh you have one kind of modeling that is trying to capture a topological description of how things and then you have this other kind of modeling where you're trying to explain why things behave in the way that they do right and that's sort of like the more dynamical systems one um so in a sense in the in the dynamical systems one it's explanatory because you're trying to explain why things behave in the way that they do and the crucial point here that we also point out in the paper is that when you explain to answer the question why you need to have a theory you need to have on a hypothesis so this means that these models in dynamical causal modelling they are hypotheses dependent right so you are looking for an explanation as opposed to a description um in uh for example structural causal modeling attracting functional connectivity for example so what i want to say is that in these two particular ways of modeling things you already find two different uh two different goals and two different strategies one is descriptive the other one is explanatory now my question is um whether modeling this uh in this way we lose any explanatory attraction i don't think we do i think that this depends on what we are doing and how we are doing because if our research is going to be driven by a hypothesis then you're going to get this explanatory traction and let me just give you a very quick example so let's say that you are modeling a pendulum right you're modeling a pendulum and you're trying to explain uh so there are two things you can do with this you can describe the pendulum and you can also try to explain come up with an explanation for why the pendulum behaves the way that it does right um and um once you do the explanations obviously it's very obvious for us because it's a very simple uh system uh the explanation is gonna be on well that's the laws of physics obviously right but you have what you've just done is you've just explained what a pendulum is doing why the pendulum is behaving like that by using these two different ways of modeling the pendulum why it behaves in the way that you do and how it behaves in the way that you do you have your why by going to by using the laws of physics you cannot say you cannot at any point say that the pendulum is representing the model and yet you explain why the pendulum is behaving the way they detect that it is so i just want to say that i'm not so sure that we should take the realist or that we need to take the realest part just on on the account that perhaps you lose explanatory attraction well we are explaining the natural world by modeling the natural world so i just wanted to add i'm gonna

retrace that gem of an argument actually realism is looking for descriptions explanations about how things really are instrumentalism is testing hypotheses and then the key twist in the discussion in the argument is our hypotheses are about explanations of the system so we actually get the benefits really without the baggage of realism because we're hypothesis driven investigators as scientists searching for explanations through hypotheses so by centering the hypothesis quest we can be instrumentalists and get the realist fruit but if we act like we're realists like we're searching for an explanation or description of a system we get lost we're not hypothesis driven and we're trapped with all this baggage of realism so very clear and really thanks for sharing it we got blue thomas stephen so i hate to like jump off of that like wonderful um explanation by ines uh like regarding realism but i have to go back to uh what scott david just said about the mind and the brain because it's something that i think about a lot and um that i am pretty deeply involved in but um language uh cannot be where the mind exists and i'm not going to say that i think that the mind exists in the brain because i don't but i think that the mind also cannot exist in language because that just it totally um you know negates the experience of someone like helen keller who didn't acquire language until she was you know well into her late childhood so so before that did she have no mind like i don't think that that's fair but also like in any kind of trauma experience people don't have for the trauma so were you mindless at that time or are do you have no mind to conceptualize the trauma so there are many experiences that happen on a very primal and very emotional level that cannot be explained or represented with language right so so i think that the the mind um can't exist there and then maybe to tie this back into the realist um the realist versus like is language real right like so so i mean i i'm one of these people that that really questions reality and what what is real like what can we take to to be a realistic representation and so language to me is an instrument so i mean it's something that like we it's it's not even it's it's a human concept so it doesn't exist outside of our perspective reality which i mean to me it totally pushes the realist debate off the table because like what even is real and i have a neuroscience background but i know like even in neuroscience there are very wide schisms like people that think that reality is a concrete objective thing but i'm like you know reality exists here and like that's it right so so i'm on that other side of the wall um anyway that's it so sorry no questions just some thoughts thank you thomas stephen and then anyone else um i think my initial point was uh just to react to ines her beautiful explanation there and i really think that should have been in the paper because it's actually a really really good way to set it up um yeah that's really good as for the the where the mind is um and what's real i think um i'd love to chat on about that for

probably hours i think it's very and i think the inactivist perspective is a very nice middle passage between there's an objective reality uh that we are approaching or it's all in the mind um however i think really getting into that will probably take us a little bit too far away from uh active inference so i'll i'll leave it at that but uh yeah the you know the conversations were on the freeway all these fun keywords and side discussions the off-ramp so thanks for sharing it steven then marco yeah i mean the culture culture does come back to how we frame looking at reality so we tend to look at reality um either with a perspective and model to get that perspective we use these tools because we're a very kind of like tool based kind of culture now um but i like in the paper it does say it says in that first abstract it remains disputed whether it's statistical models are scientific tools but then it says to describe non-equilibrium's steady-state systems so in a way non-equilibrium steady-state systems is kind of like a foundational ground that that that could be to some extent real it's the question of the models that are used to describe how they work that isn't real you know so i think that's also like this doesn't like there's still room for the idea that the brain at some level even though it's quite abstract is a non-equilibrium steady state system it's just that the models the statistical models shouldn't also be the i don't know the way that that system's doing its job so to speak so i think that's kind of and i suppose what that also then brings back is how do we get at that and this is where i'm quite interested in indigenous ways of knowing because well if you don't come from a perspectival root and if most of what if you come from other like metaphors is one level but other sort of swarming kind of um non-non-literal ways of knowing you know non-verbal maybe you that that's another avenue that opens up that says okay maybe we have to come at it through other ways of modeling anyway thanks steven marco than anyone else hi thanks um so first of all uh just fair distortion i haven't fully read it yet um but um it's an initial discussion and and uh scott mentioned paradox and i'll take that up um i actually suspect that um the paradoxical interpretation might actually be the correct interpretation so we have a nice um table of possible interpretations for the fvp um a two by two uh table but but what if it's all of them right so so the fep especially in the elaborations and active inference is already kind of paradox because you already have these trade-offs between accuracy complexity or the epistemic and the pragmatic aspects so in kind of a weird way the whole discussion can flip up your other different interpretations seems to me to parallel the way that the let's say first deflationary the statistical densities in the brain actually fulfill different roles depending on the actual relation with the world right when i'm actually just imagining about the world then the representative relation between the densities and the targets are not actually with respect to the

world they're actually respect to each other in my brain right but nevertheless uh we still situate in the real world at the very least if we assume an objective reality as in a reality that is not dependent upon our interpretation until we act upon it then at least we can say that by virtue of fep at least what it suggests that the way that these models sorry the way that these statistical structures and densities are shaped are shapen by something that is so in that sense i i personally um sorry i have a lot of allergies for for some stuff philosophy but um i i'm very much against this whole notion that the representational content or representations have to be adjudicated by some accuracy or correctness condition and i think it's exactly fp that challenges that right why the hell would that be reasonable that a reputation has to be correct rather it is that it is shaped by some sort of correctness right but if you would be stuck in that we would never get somewhere right as an organism it's not about being true or wrong to correct all the time it's about being so ratchet yourself in virtue of this adaptive imperative um i'm not sure where i'm going um i still have to properly observe everything but um maybe not actually type this up is what would you think of um instead of choosing one of the tables um cells one of these possibilities more embracing what is in my opinion applying the fap and saying that it's actually paradox if it's all of them and none of them thanks marco and i also agree the fep helps us by thinking about how the observations get to us whatever side of the blanket or the interface we're on we can take those observations with deathly seriousness instrumentally it's not that the actual movement of the water molecules is gonna kill you but there are temperatures that kill you and so we can take our measurements seriously and be instrumentalist from our side of the blanket and say yeah you know there might be a real system let's hypothesize as instrumentalists about what is out there but if we're waiting on our side of the blanket until we we have resolved what is quite literally on the other side of the blanket then it's going to be arguing about what's inside of a box that can't be opened and then policy failure is going to be the outcome so it's a really interesting point about how we're kind of using this fep scaffold to actually navigate this paradox so really cool there so annes scott thomas um okay i just wanted to add to uh what marco was saying i think it was a yeah it was a wonderful uh remark and i just wanted to because that also allows me to sort of like draw to highlight a point that we made in the paper that was also very fundamental and this is just that the fep in itself does not have to entail this table we can talk about the fep without talking about this table and that's a point that we try to make in the paper so this table only comes up as something that is relevant to us once we start talking about process theories or the corollary uh active inference because the fe p in itself uh comes from uh variational free energy which is

a measure of the relational variation of uh free energy to attain model evidence in relational base so we don't have to talk about this table to talk about the fep okay and we don't even have to active inference or process theories that's when we start talking about okay how is it that the system is minimizing free energy that's when we come up with these process theories to try to explain how is it that the system minimizes this free energy right and that's when we start bumping into these conceptualizations that um our explanations or our models that aim to explain the process by which the systems minimize free energy also exist in brain right so we don't have to take any really standard or any instrumentalist stand until we start talking about these process theories and i think that sometimes i think that it happens a lot that in the in the literature these things come conflated that the free energy principle necessarily entails active inference or necessarily entails or is compatible with process theories like prediction error minimization or predictive coding or protective process process and these are actually not there's no entailment between them as as we saw and as we as we try to point out in the paper if i could ask a question quick follow-up on that actually so let's say we're in the fep world then there's all these corollary process theories that are falsifiable we've talked about that um last week and in several other weeks what is it though that leads someone to accept or reject or be attracted or repulsed by the fep so within the fep we we know how to get from here to there and you're right there's no strict entailment it's actually a totally scientific empirical question but how do we get to that fep understanding that leads us to the downstream process theories um okay um i think that are i think that there are different goals and then at least for me it's useful for me to to to think about it like that so what we have what we gain we're talking about fep is that it gives us a much more sort of like a grasp or gets us closer to what things really are biologically speaking so what we get is we get these um these bounds between life and death by the form of entropy so it is this key concept which is entropy that you do not get in process theories because precisely process theories are in a different game they are in the game of uh explaining uh the processes by which uh cognition is is an activity uh uh is conducted so it's the processes and the fep is on a different business which is the business of looking into stage that's why it comes much more related to dynamical systems theory so that's the point that we tried to make uh clear in the paper that it seems to us that comes very often conflated in the literature that there is an entailment i mean active inference is a corollary of the free energy principle so it is it is very important um precisely to tell us how it is that systems could be but there is no necessarily entailment even though if you want to tell a much more detailed story then active inference becomes very useful thank you scott

thomas anyone else who raises their hand thomas i think you had your hand up first is this something you want to jump in on no you okay you're meter thomas ah yeah my fault um unless you have something that's um not really related to the current conversation uh you can go first because um if it's related if it's related thomas go for it otherwise yeah yeah go for it um that i i noted some comments to a lot of the things people said uh um so regarding the question of what what we get from accepting the free um i'm honestly not very sure i'm still trying to figure that out and i'm i'm hoping we can get some very neat statistical models of behavioral dynamics and that we can somehow get a better grip on the patterns that we see um and then coming back to uh what uh marco said um if i understood um the story correctly by saying that um you know why not all of the options simultaneously right so in some ways the model is a very neat description in another the brain's model captures something is based on something real in the world um i think what happens there and this is this is something that i've seen in the literature as well um i don't remember who wrote about that but i've seen some people saying oh well you know it's the best of both worlds because we get both representationalism and activism um and i think the issue with that is that it assumes that realism is a valid position it assumes that we can talk about such a thing as a model in the brain in a non-metaphorical sense it assumes that there is a model in the which does or does not latch onto something real and i think in that sense um the way that you often see people talking in activism and the free together is by assuming realism and um ignoring the inactive criticisms of what realism entails um and it was the second point about whether we need to see representations as involving non evolving correctness conditions um i think that's a very fair point but my response to that would be that sort of comebacks comes back to our um talk about encoding decoding and language and being clear in general um we for the sake of our paper decided what we call representations involves correctness conditions um so if you have a different notion of that does not involve correctness conditions that's a totally different story and maybe those types of representations could be totally fine if all you mean with representation is a covariation relation between two different systems i'm not going to dispute whether that exists or not that seems fine but the problem is that if you are that liberal with what representations mean um everyone's going to use them in different ways and you're going to get that we all use them but nobody really wants to talk about what they really mean and that's pretty pretty famously at least among philosophers described by bill ramsey in a 2007 book representations reconsidered i think it's called um so the reason to to say representations have correctness conditions is mostly just to get clear on the terminology right so just a signal

of this is what we mean when we say representation and if your notion of representation does not have um that's something else to talk about that's the another discussion is whether those are valid or not then thanks thomas and in an earlier discussion with um maxwell ramstad and others we had a conversation about how part of representations is that like they have to be able to be wrong or they have to be able to be off base otherwise they can't be on base we've so yes lots of other interesting work um scott if you would like and then stephen marco just said something going back to the earlier conversation it was when we were talking about the um the basically the again the ism schism and the ability to i'm staying with that and the ability to see inside another system and discern what was going on in the system it reminded me of the holographic principle and i started reading a paper another paper last night on why holography and it was just kind of an interesting notion that the markov blankets and the holographic principle bear a similarity in there's an opacity to them and so the internal states and the perception of those internal states from outside may be based on different isms and that's something that may not ever be um bridgeable i guess and it may not matter we're gonna talk to chris fields about holographs in march 2021 right so coming up very soon thanks for this great point scott steven and marco yeah just to build on the um this idea that you go from free energy principle and then we go to this process theory so it's almost like at some point you've got to put dimensions on what's going on like there's an inside there's an outside there's so with active inference there's there's the idea that the stuff is going to go via the markov blanket and there's a way to know and that way of knowing still cut it's still to some extent dimensionless because um the there's there's no direct way to transmit the dimensional information it's only through this entropy and i know entropy is mentioned in the paper so i'm kind of interested in this idea of when you go from say the entropy in this kind of um general sort of non-dimensional sense it's just it's just a way of knowing um and then you've got this kind of high you talk about this idea of a low entropy state or high entry states but you've got this kind of this is sort of the jump between the two in a way where there's like how because with active inference it's the change in entropy which gives the way through the back door to encode knowledge so because it's through the change in entropy in relation to action in relation to sensorial data that you sort of somehow build up a dimensional understanding even if that understanding is only in the mind maybe not at the organism level so i suppose um i i think that may be what i know there's a new paper you brought out on skillful performance has just come out so maybe this is sort of opening up these other questions around entropy and dimensionality and when you actually go from modeling changes to how do these changes

relate to information or knowledge flowing or being generated you know from the blanket through to the generative model and the world yep stephen thanks a great point you're getting at this question which is how does the dimensionally compressed information from the world like a panorama that just presents itself get enriched through our experience and internal model into something that has what you're talking about is dimensionality and then it's a good question to ask how it relates to the thermostatistical lineage with entropy and some of these very pervasive top-level concepts so really well brought out marco um uh are you wondering um about about thomas's comments on my comments um if you could elaborate on first of all the um uh the criticism from an activism as he mentioned but but i would love to hear more about that especially in relation to um maxwell rancid another paper on an active tale of two densities um also about the correctness condition i understand that you have to choose a definition of representation and again i do understand that you have to respect the lineage in philosophy and what has been established as authoritative or conventional um but but um i i do believe that that there is a certain need i think to also revamp philosophy because i feel like there's always this kind of weird dynamic of like the two clubs in the school playground with euphrosion science one is constantly trying to radically revolutionize the but philosophy seems to be so stubborn in not allowing science to reflect to revolutionize them right they keep the philosophers keep trying to you know to to to yank the rigid price out of the shell but vice versa somehow there seems to be no path we're not allowed to do that so so for example why not let's this field of science philosophy um gank that prior of representation having to have correctness conditions out of that shell because for example you can you can also just kind of play with it right not to correct his condition but uh saying that in the limit it should settle into correctness relation with the target as if it's kind of fixed point or attractive point right you can also say it's not a condition it's not a limit but it's a factor that contributes to that thing that we call representation right so so i think i would love to hear a discussion about that opening up the whole role rather than this essential property the role of the notion of correctness in relation to representations as situated in the world um and in turn i also want to again emphasize so even though that you're right in the difference between the coroner the many corollaries potential corollaries of fep versus fp itself where fap constrains a particular way we create um and there's another thing that i think is not often enough talked about which is the kind of implied picture of sentient systems for uh from fpp and its corollaries where we not often enough talk about the inter-relational um system the way that the systems incentive agents or complex systems relate to each other so it's more specifically for humans or us um

we also self-model right and when we're self-modeling we need to model all the different structures we possess and know what role they play in the big ecosystem that we are and so in the same way we can say that we in our self-modeling we also intuit or sense that some of those subsystems have a very instrumental right if for example you could say that the executive or attentional mechanisms of the prefrontal cortex is effectively instrumental and in a way realistic about itself but the fact that it has to organize regulate itself such that non let's say primary physical observables needs to remain a certain bounds that seems to me instrumental because it doesn't matter if there really is some some quantity or some notion of stability of feeling good or feeling safe that doesn't matter right it's just settled in that way but at the same time if it does latch onto something real cool and maybe you can't elaborate upon it sorry yeah yeah it's all it's all good no you're bringing up okay so point one was related to inactive critiques point two was about correctness conditions moving beyond them point three is sentient systems so if we want a question that's answered let's just try to trace out the threads so that we can actually return to it because they're great points yeah it's great points marco is that good or is there more it's just what's the new ones the third point is more about in the process of self modeling we include the notion or or implicitly intuitive model of the notion of do these things have instrumental role or more of an epistemic realism right that that that way more that these theories are more about modeling processes of modeling rather than modeling essences and properties yes the relational all the way down type thinking so thanks marco really three sides of a very nuanced discussion and an ongoing questions so thomas than anyone else who wants to raise your hand yeah thanks for that um so i'm not sure i completely follow everything so um if i get it wrong just interrupt me and um let me know um so i think you mentioned um regulation like bodily regulation as a form of self-modeling right um so i think that accepting that bodily regulation is a form of self-modeling is assuming the realist position the there is a clear effect that we see biological organisms self-regulate whether they do that by exploiting an internal model is another question entirely and that's exactly sort of the distinction between instrumentalism and realism that the paper tries to tease apart so it's almost that when you mentioned the um modeling the process of modeling is the instrumentalist side on marco's way of phrasing the third part and then the question about essences is the question so the movement towards instrumentalism is related to this relational insight so i think the issue is i didn't exactly get the relational insight that's the part that was a little fuzzy for me and i wasn't sure exactly what it was getting to yeah marco where do you see that relational just because it's something that comes up and and um it does sort of cross

different idea or debates to so what does that mean to you or how does it play out here yeah so thanks so so often when we when you see literature on academics or friendship uh there's always a centerpiece of a market centerpiece of a certain system slash agents but the very rare one figure one i'm showing figure one just if you want to refer to this oh yeah exactly so it's always this but it's never this with another one that's like this it's very rare let me just see market blankets uh supplies or markup lengths of live paper that that does go into it but but very little and start the shockingly like these kind of philosophical papers um is there actually talk about um how market blankets interact with each other and the reflective of the mutual influence because in that process of mutual influence how do you actually do that right how do you model the other models that you possess or are right you're not just it's not just in terms of properties as such it's also about the role they have the bigger inferential ecosystem but i guess maybe that that requires a leap of faith in active influence first right if you already assume that the dynamics are governed by inferential imperatives then you also have a small step to believing that in modeling each other they also need to model each other's inferential contributions which where we go again back to the decomposition of free energy principle uh formulation in terms of epistemic and pragmatic or the accuracy of complex right maybe one mode of engaging with or incorporating one particular set of systems of your total constitution will lead to more instrumentalist approach right when you're just um instrumentally engaging with something not assuming you have a correct view of something when you have a new game or some weird object you don't know what the hell it you're not gonna engage with it based on a realist assumption you're gonna engage with it based on idea of i'm gonna figure out how to manipulate it i'm going to find all the instrumental points of latching onto it right and so no matter what you you cannot escape the the fuzziness or the vagueness or the fluidity of these so seeming dichotomies in my opinion and that's the relational aspect right that in terms of modeling and its many facets you need to understand the relations of these different systems as yourself model especially if you're extremely complex with humans playfulness human as a curious investigator playful insight relational thinking and a very something that kind of is personal and almost distinct from philosophy and or science it's the personal which is we're really studying this and we're people who have certain um valences towards different modes so pretty interesting um thomas in us yeah um so i think i think that's a very question um and whether we approach something from an instrumentalist or realist perspective um but i think still that all i mean i might be way off base here but from what i hear it sounds like it's all still within the realm of realism and only when you take us to be modeling at all can you say

about can you talk start talking about whether that modeling is instrumental is the realist so the instrumentalist position that um we defend in the paper is one in which we only model when we literally do so in the world when you're using your computer ram to start up a model or when you're drawing a diagram of when you're literally making models in any other sort of activity that you're doing you're not you're doing the activity so in picking up a cup i am not modeling the cup when i'm not in any sense models are not relevant there i'm picking up a cup i think the other point you made a little earlier about the interaction of science and philosophy and that it seems that philosophers are always trying to revolutionize science and science is not allowed to revolutionize philosophy um i think that's a fun point as a philosopher um but i think that i think that with the popularity of predictive processing and active inference in philosophy you see where science really is revolutionizing philosophy as well and i think especially in philosophy of cognitive science there's always going to be a bidirectional interplay um i wouldn't have been sitting here if it were not for the influence that science is having on philosophy of course i'm delving with into all of this um and then when you're talking specifically about the notion of if scientific endeavors could make it extremely plausible to throw away the correctness conditions for uh representations then maybe that is something we need to accept but the issue is that i think anyway just at first glance so representations are it's a word we use also outside of science right a picture that i draw as a representation of whatever it is a fancy drawing and there are certain features of the picture that i drew that are particularly relevant in making that picture a representation of whatever it is i drew now when you want to use that same word in science you might just say well none of that is important and we're going to drop some of the very important features of the picture and use it very differently in our science jargon that's fine but it's a little difficult to deal with um when we already have a notion of representation and what representations are in um just the regular dealings in the world but you could technically say sure you know we'll call it something else but it's something that we'll have to all agree on and then be clear about what that means or not thank you thomas so s and then anyone else so we have like little you know half an hour or so left so keep any thoughts prepare them get ready we can flip around to different slides in s then markup um yeah i just wanted to uh i first wanted to say the great points uh thomas uh yeah uh that's precisely what i'm gonna sort of like uh expand or try to explain a little bit on there's there's a a lot that that i'd like to say but let me just um go back to the self-modeling point as the bodily regulation so i'm really struggling to to understand what this means and the reason for that is because we need to motivate why would the body have to

model itself so typically one starts this modeling um sort of like um endeavor when we do not have direct access to things so we wonder um what could be you know causes so i wonder why i'm really struggling to understand what this self modeling is because uh in terms of body bodily regulation because what is it that the body needs to model really because um we start modeling we want to model things when we don't have access to a true posterior when we do not have the causes so we need to find the posterior right but the so the self modeling seems like a little bit counterintuitive because it's it's the self that is right there so the body is right there so it's it sounds very counterintuitive to me to that that should need any kind of like internal modeling um in in in the body um so the other point that i wanted to uh get back to uh was marco's point because marco seems to be like a little bit like worried about um you know the lack if i understood it correctly the lack of um having sort of like a a program or a framework that would allow us to get like things um an explanation there would be like much more relational all the way down um and how would we like link all of this together and then that's what when we came to to this slide i think so i just wanted to say like a few words on that i want to say that the okay so that the cool thing about markov blankets as we know is that uh well we can apply uh regardless of the scale you can apply them to any scale and you can nest them and i think that this is something that um marco could be like a little bit worried about and perhaps this is not going to be you know the solution for what it was it was mentioning but maybe to get us like stuff in the direction so the idea is that um with markov blankets we um actually understand the relations between different aspects and levels of uh neurophysiology um as being uh uh together relate related in a multi-scale kind of system so basically the idea is that you can use a markov blanket to zoom into the level of description that you want to investigate but without losing sight that this particular level is already embedded in a multi-scale level uh system right so then this allows us to uh zoom in um up and down for example within the brain and uh that's a paper that i have um is going to be out soon so it allows us to zoom in and to zoom up and down in the brain but it also allows us to zoom uh out to the organism and then get also the organism um in the environment so that's the a nice thing that we can do with the with the markov blankets where obviously the internal states and the active states are going to correspond to the level of description that we want to look at and then obviously we are going to look at the embeddedness of this system within its environment because its environment is going to be the external states and the sensory states so that's the cool thing that we can do with markov blankets is that it allows us to zoom in and out um um and up and down in a multi-scale uh thanks yes great points and i think for this

zooming in and out seems very relevant to a lot of our work and a lot of what's in the cutting edge so maybe we can have like a workshop or panel or something like that or some we'll see it later but this is really interesting stuff so again maybe you know 20 30 minutes so if anyone has live chat comments marco then anyone else who uh then stephen and then anyone else who raises their hand and then scott thanks um so she want to go back to uh models and multiple levels of skills and stuff so so um uh thomas says something about you're not modeling uh something you're just doing right um and i agree again like i said uh my my claim is that that that it varies over the situation context so again that's why i argue that you need to zoom out to taking the relation so if you take the paper um on active infinite curiosity with the with the games right so first you're exploring you're just doing the game you're kind of allowing the statistical patterns as are impinged upon you to kind of accumulate shape and adjust each other etc and indeed at that point you're just doing and you're just at the emergency co-variation of relations etc but at a certain point there's some kind of transition you have a moment of insight the model sorry the let's say the picture more naive because in sense of the game kind of arises into your awareness as a complete picture and at that point you're able to at least hypothetically write down the model that you think that the game is governed by right and so you said something like we're only modeling when we're doing the computer thing and claiming that this is going to be a rotational model of the environment but the thing is what what's the chicken in the egg right if you're doing that game you're at the mercy of the patterns and then you obtain this internal understanding of how the model looks that allows you to create a computational model that means that the point where the model is substantiated is actually in the agent itself right and so that's why i talk about the relational thing and the and the sound regulation when you add a system are able to model yourself then you have different orders of that model the moment that it's just intuition just you're guided by the variational patterns just letting that be latched onto your actions and policy selection yeah it's not really maybe just maybe it's not the model then but the moment at this level above you're able to say that this is a model of that and this is the the systemic relation pragmatic relation then i think we should be able to say that's proper and model proper right and another thing so something thomas said about uh when science allows us to throw away the correctness condition i think we have that in psychology and in social as social psychology a lot of people can develop an understanding or a model or a stage of the world that is actually not governed by correct conditions if you i don't get political but you know you see in certain political areas where people seem to be governed not by truth conditions but more about pragmatic conditions right

but which beliefs if adopted or which ideologies if adopted will i be better off in the world right that is not a correctness condition that's that is taking instrumentalist or pragmatic primacy for the sake of your fundamental imperative of autocracy self-preservation etc um and so and and and and uh in this uh point about non-direct access so yeah we're oneself but it's most level motor skill so there remains inferential distance between the locus of certain areas of yourself and other loca of yourself so the hypothalamus is crucial in in self-regulation in terms of guiding the endocrine system but obviously it's distant from the actual systems that is supposed in relation to right so there it needs to be an acknowledgement of the inferential relations and infrastructure distances by which the self as a whole learns to um uh yeah i just leave it at that great thank you so in the last little bit let's think about tying threads up and phrasing things as what we wanted no it's great marco you did good and exploring what we want to do next week so we're raising questions to think and for next week as we tie up some of these really awesome threads so stephen scott in s thomas nice um so well i just wanted to sort of add on to this um or sort of maybe this is something that could come back to um this idea of representation i suppose it is ultimately seen as something that's in the brain and a model is also sometimes this is maybe where we get a problem is we kind of have this folk psychology that models are in the brain as well so if i think if we were to say models are in our niche so if we say we take this figure one the model is something we create in our external states in our niche that we work with and i have tools and we manipulate out there and we engage with our active state sensory states internal states to work with um it can get rid of uh it helps to sort of um get around some of these challenges um and i think because i think that that is um the nature i think this is really useful thinking about correctness and accuracy of a model is when can that need for directness and accuracy be um loosen someone um because it might be that you don't have to model the body doesn't have to model its active states or model its sensory states or model those exactly but it's somehow using something like them and there's some sort of bigger model which we can try to use as a tool to understand what that aggregates to so i this is actually maybe opening up quite rich thread thanks steven scott and s thomas shannon i unraised skip excuse me thank you scott s thomas shannon okay so um i just um i think that uh it's important the question that um marco was rating so i just wanted to sort of highlight that and just like sort of gesture to it for the next session um so i think that uh marco was mentioning um this important self-regulation aspect um and i wonder and this is like a very embodied kind of thing it's not you know available to our conscious level so we don't know what's happening so it's not available to conscious consciousness and i was trying

to dispute that idea that it needs to be to entail some kind of like modeling aspects because for us to you know for you to tell modeling aspects it's because there is not a direct access to the body and i think that one follow-up question to that would be to think about what self-modeling would mean on a conscious level do we ever do that do we need to self-model on a conscious level right because if we do then perhaps there is like a sort of like a barrier the self and what i think so it's kind of weird so i would really like to discuss that um with everyone and um yeah and the other question that i think that would be nice to discuss next week is to actually get down to the nitty-gritty to the detailed technical details um that would allow us potentially to think these things are what thomas and i had in mind in the paper which is whether from the fact that we can be trained uh to use for example variational uh base to uh attain likelihood of uh model evidence which is like what we do vision inference and modeling that kind of thing and also from the fact that we as human beings in our daily life we can think and engage in sort of like abductive reasoning to wonder and make inferences about what what we think is the most likely case of things around us from the fact that we as scientists can do that and we as you know regular people in daily life can do that does it follow that the nervous system itself or the body itself does that and engage in those kind of like tools so those would be mine yeah nice i'm already capturing them on a slide so we're definitely going to get to them ines thomas shannon marco oh sorry yeah thomas shannon marco yeah thanks i have a lot in answer uh to what marco was saying so i'll try to slim that down um as a very quick probably not to um convince you but just as a description of how i would see things um if you're playing a game and at some point you figured out the patterns of the game and you're able to make a model of it i would say where the model is instantiated is when you draw the model out that drawing you made the description of the patterns is the model and when you're producing that drawing that you can share with other people that other people have access to you are modeling um any other modeling models are never in models are never in the head models are in the world in our interaction with the world and only exist because of our social cultural heritage in which we have learned to use models sort of what ines was getting to as well um and then when you say when you're some people being led not by correctness conditions um i think that's that's very fair and i none of us are generally led by correctness conditions in most of what we do except for science when we give it our best shot um we're just doing stuff but the most important thing there is not whether that means then that we need to change representation for it to mean something not involving correctness but maybe that means that the notion of isn't really applicable here and instead of changing the

words we use we just use a word that is more appropriate to what people are really latching on to and i think that'll be just communicatively maybe an easier strategy instead of continuously changing the word to fit our current understanding and just sort of come up with a different word that gets better some deo logisms and also our growing terms list shannon then um thanks for all of that discussion i'm wondering if in search of terms that would better fit our description is the term expectation more or less vague than saying representation or a model and what i'm thinking of is when you're standing your body has a certain expectation that you'll remain standing if you're standing in a room you expect like you at a conscious level maybe expect to stay standing but also um you know your cerebellum and muscles and your trunk are expecting to be in a certain relation to each other because the ground's not moving so they're just maintaining you know some standard standing posture maybe a little bit of balance side to side if you feel like you're standing still on a train and then the train starts moving when your expectation was to stand up straight now you've have a deviation from that expectation your cerebellum your motor cortex all of the muscles in your trunk and your legs react or predict that they should have been standing upright predict that they should have been in a certain relation to each other stand up right and they either update that prediction or they act to change the world so in if it acting to change the world then all of your muscles will tense up in a different way to keep yourself standing upright if they're just accepting the prediction that you shouldn't be standing already you'll fall over and i just wonder is it helpful to say that your body has a model of standing or like an s says your body just is in direct access to itself so even if there's an expectation of there's still like maybe it doesn't help to call it a model but there could be correctness conditions in that case if you called it a model and like if you're so far out if you're so far out of allostasis right then you're in an incorrect correct correctness condition for and so to respond to that perturbation is to bring it back to that that correctness condition and i don't know i don't know i guess does expectation help or is that just even more vague than saying that you have your body has a model of itself fun thanks shannon uh marco and s yeah actually um what channel says very nice because i think again that re-emphasizes the importance of of distinguishing the kind of operational constituents of how models play a role in agents so what she was describing seems to me like the interaction of of um how the uh afferent and efferent expectations and data have to interact and that indeed i agree that that doesn't have to be called a model i would just see that as um the element or the operation of inferential interactions right and and kind of a zeroth order or a proto model maybe but again um it's on top of that when you have the kind of order above that where i think you would

have to say that there is a model because then there's also the modeling of how these infrared constituents relate to the world that has the model be modeled too uh for example sometimes we interact with the world in weird ways and we notice it and we have to actively reflect about how we were guided by our actions i mean i don't know which philosopher was but you know philosophy about self examination examining your beliefs assumption if that's not modeling i don't know what it um and and um he has raised some very nice points um so the rule of consciousness uh too about conscious of regulation and i i personally believe i am one of those people does attribute a function to consciousness or at least attribute a functional drive that led to the emergence of consciousness where by the sheer fact that it is uh the kind of the apex of higher and higher levels of modeling similarly how multi-sensory integration is a consequence of this in the very same way um it seems it seems stands to reason especially if you look at works like graziano's attention schema modeling that's the very drive that led to motorcycle integration and consciousness is the need to model that which is below that um i'm sure there's some there's some activation papers for that too um i forgot in response to what it was but also wrote down somewhere that um it's relevant to take also mel's paper in this discussion about models because i think what's very beautiful there in contrast to this discussion is diversity of nuances regarding what a model can be and what they are useful right although that would probably stand at us with a lot of inactivists but you can draw a lot of parallels between the different roles and values that model could play and have for a scientist and the various ways that models as embodied could be useful right for example some are good for for conceptual exploration some more truth-oriented some are more to kind of integrate different things and i think these are all valid ways to look at models um and they survive simply by virtue again by virtue of what free energy principle entails is that the the success of these models is what leads them to persist right and the success in terms of their pragmatic and epistemic contributions to minimizing or reducing variation free energy right and so it's like oh wait yeah alex uh john says a recent paper right or well one of them had the recent paper on it's not about it it's about uh what must be true if you've gotten there actually right so what must be true yes yeah yeah x constant so so given that we have bernoulli assume models what must be true uh given that we have them right so the answer would be the uh they got there because apparently they contributed to a reduction of variation free energy um also i thomas i gotta say i think it's it's kind of one of those uh disagree uh agree to disagree things because i i i i feel a sense a bit of an axiomatic commitment to models simply cannot be embodied um although i would love to continue that discussion and i also think that adapting a language is key and again taking the

the norm of principle free energy it's the philosophers who disagree most people use it like the way that i've been describing right folks psychology and scientists use in less of a seemingly anal status rigorous notion of correctness condition it is simply a part of a bigger picture how their value should be assessed so i i and i think in general it's a it's example of niche construction we are embedded in a world of language we're our linguistic bodies so adapting a language instead of throwing it away seems to me a far more reasonable approach um especially if you want to confu if you want to avoid confusion so yeah i'll leave it well it's always interesting to re-listen to one's live stream because while we're speaking we think a certain way it's a different mode and um and s you really have a finesse with constructing philosophical arguments on the fly as many other people embody and the way that that is done performatively stephen i'm sure also you know resonates but like the way it's done for the first time performatively is very different than hearing it in a different state of mind it's like it lines up in the moment but then a different moment and then it's like because that's so it's really interesting points like are we self when you're improv when you're improvising are you self-modeling or are you doing are you just controlling your larynx where is consciousness fitting into all of that those are really great questions in s and then any like closing thoughts we could go anyone else who wants to speak in us and then feel free to drop off whenever you want um okay i just wanted to sort of like address uh marcos and then address uh shannon um point very good points um so there was one thing that marcus said that uh strike me very very interesting because you were saying um about that we need to so you need to choose the models the models that are more um you know suitable um and i think that um by saying that obviously we're taking a realist a a realist perspective as in thinking that there are actually those models in the body or in the brain so that's sort of like what thomas and i challenge so i just want to sort of like give my two cents on that particular thought and say that i don't think the success in is choosing um in choosing models but it's more like the success is in choosing the most desired states that are going to get you what you want which is like you know avoidance of dissipation entropy uncertainty and all of that um so states are preferable models don't exist so that's kind of like my take on that and then i just wanted to also address what shannon was saying which was really interesting because i thought that it seemed to me that channel was trying to you know build up like in okay let's just really get down to hands-on and see like how do we really apply these things that we are discussing here into like for example my lab um and i really like that um one thing that i think that is interesting that happens um uh that i see happening a lot with my conversations is that we don't even see how we are already

sort of like in completely and fully enculturated in a way that we speak about the body and the brain and cognition in general with so many assumptions that these things exist because it's just the way that we've been enculturated um so as uh shannon was describing oh i think that wonder if maybe its expectation is a better word right and then like it strike me because i was thinking about it okay um so the body has got this expectation but from what from what so is that is there like because for the body to have a certain sort of expectation i would probably think that maybe the body would need to be like sort of like separated from the brain so it had what would have like some kind of expectation um or um uh some kind of like inbuilt system where the the body becomes like sort of a humunculus and also has its own expectations so detached i i i was like a little bit like i'm not sure uh what to do um with that um so um yeah so that that would be like a point on on whether the body's got this expectation and um and um and and and how i think that it another way to think that would be to the whole organism is evolving towards right there's a sort of like at least like one goal that is shared by all living systems right of course interactions are very idiosyncratic we're really context dependent but there's like one goal there's like sort of like in general times shared by all living systems is that they don't want to die so the the so it is evolving towards uh sort of like avoiding this dissipation so whatever whatever the state the body is at it's it's got to it's got to be dealing with uncertainty right either so then shannon was giving this the example of like oh it's standing so so is it there's an expectation of like a certain expectation regardless to in regard to being um standing i think it's more of like the state that the the organism itself is which is either certain or uncertain it's stable or is not stable it's it's it's in in in it's it's it's fulfilling sort of like a a state that is going to you know lead me to to the next stage so i think that would be like a a a sort of better way to describe the behavior um that is entailed in in yeah in this cognitive activities that we do because i'm not sure how to think like the body itself expecting something but maybe i'm just i i missed um i missed something here bill thanks so much for that shannon marco and then anyone else who wants to give the last thought great thanks for that ness i think um where you sort of ended where you have like most organisms have an expectation of being in a more stable state than a more unstable state depending on the context and depending on the environment and everything um but the point you made about enculturation like how does the body have an expectations um and i guess when i said the body i kind of was implicitly including the the brain so if when your body's actually upright your brain and your muscular system have an expectation of being stable in that upright position and i think enculturation um we could also just talk about um like the development of

certain structures or expectations or situatedness or something so the same way like enculturation gives us an expectation for a certain form of language our body just by existing in the world and being in standing states in lots of different develops an expectation of what a stable standing state would be like um but i think where you ended sort of on this like constant evolution sort of away from death or away from perturbation or towards stability i that covers and that was helpful thank you shane marco and then any other comments thanks um regarding anticipation in the body um some examples i'm not sure if this translate is normally dutch but you know if you've heard of it or or pork soup is eaten with a spoon but most people will say fork especially when you're kids in kindergarten so um that is more of a linguistic example because it relies on a notion of rhyme which you feel to it and the prediction error is again contingent upon the model or the linguistic policy you have um but also just in in general um for example with brain relations or i don't know imagine you're at a bar someone really drunk comes to you and there's a really big guy with scary tattoos and there's no words need to be exchanged but if he comes really close to you grabs you tightly in the shoulder there's gonna be expectations and it's felt in the body and a lot of these expectations especially imagine you're a racist you know and and someone colored comes to you so then you will feel a physical response of anticipating a certain action or something but because it's a contingent of uncertain racist beliefs which in culture there clearly needs to be a certain interrelation between the models with abstractions that you've gathered through culture and the physically or viscerally felt anticipation of an action and so i think in a lot of ways what i'm trying to say is that a lot of the physically embodied anticipations clearly are contingent upon abstract or model or representations of something because if not we wouldn't have these visceral bodies so i think a lot of my personal qualms and activism is often they look at the manifestations the tip of the iceberg the marks the labels but but not the necessary path that it has to have taken to get to that manifestation um and you said also about you know we don't need models we just get to our preferred states but then you get the problem of how do you get to your preferred states it's not that easy i want a million dollars and a happy family but you know how do i get there and so it's gonna be contingent on a lot of complicated detours and i simply don't believe that you can escape the notion of a model if you let go of the whole um correct this condition but i i don't think you can escape at least a model in the broadest sense as nell's paper i think nicely alludes to especially with these examples of getting to preferred states given the complexity of our worlds and given the cultured reality of that experience thanks mark so i would love to talk about that next time yeah awesome well on discord or

however we can continue some text brewing and thinking and it's awesome because it's just like a snapshot into our learning and thought on all these very important questions so stephen scott in us thanks well i suppose one thing in this mention there and this is this is quite rich is maybe how does expectations can expectations always be and to some extent they can't by minimizing uncertainty it's like a backdoor way to get is it you know so it's kind of um but i do think there's something quite interesting about um another way of thinking about beyond the model and beyond some internal brain state and that's where not but there's something called mental space psychology which is quite interesting about how we might be constructing and designing um understandings of a space now i'm not going to mention that now but that that something that comes up is is there's maybe there's other mechanisms for modeling that we with which is in between just the body moving and it being in the brain but is is everything tied to this active state sensory states relationship so it has to be capitulated at some level using our sensor and using our actions so there has so anything that's represented in a way we think it is somehow being capitulated in this active inference process it's not being just happened offline so to speak without any kind of embodiment yeah thanks thanks stephen and the good news is you only have to mention the keyword wants for the ai so scott and then in s one thing i was thinking about is you know we talked about this as a way uh method of analysis of existing systems biological systems etc and it's also uh the thing that's what i've been doing historically is creating synthetic markov blankets and i think i alluded that that before a contract is essentially a synthetic markov blanket that creates a being pursuant to the attributes in the contract and so you're taking two different systems and linking them harmonically coupling them through a set of expectations that are documented in the contract and so one of the things that's kind of interesting to me is that idea of realism versus instrumentalism i think it goes away in the context of synthetic systems because you're essentially instrumenting a system from whole cloth saying hey we're going to correct create this entity and it's going to have these attributes like a corporation doesn't exist except through the corporate law and the attributes that are assigned to it by shared meaning but it's all fiction but it's a very broadly shared fiction and so it's interesting that notion of what we use as descriptors for existing living forms and those things that we set up to create niches in a sense for the new synthetic forms and that's something that i find intriguing with this model from the paper is that it's when it's taken to the synthesis of new forms it changes i think it changes that analysis because it's not a question of discovery but of creativity in terms of the setting up of the attributes in the markov blanket scott if i could rephrase that it'd be like the realist is

focused on those essences of the world they're going to try and describe what's already out there that's like big databases of our current legal systems but once we're in instrument landia we're talking about designing apparently arbitrary fictions and then manifesting them and enacting them and so there the rate limiting step once you're taking the instrumentalist turn is literally the innovation and the narrative build up and how much collaboration you can achieve pragmatically without getting into the realist allowing people to have a different take on what is quote really so really interesting and it may allow for the design of scale-free interventions in that case right because you're not just relying on what happened naturally but rather you can create some predictive possibilities by the way you design one level to allow for the engagement on a scale freeway thanks all right ns and then all the last comment okay so i know that we are already a little bit over time and i do have another meeting so i'm just going to try to be like super concise and i just want to say like one final remark regarding regarding the realism instrumentalism so uh i just want to say that um imagine that i have a model i just have built the model to understand or make sense of brain data for example that i collected from the scan um i can have two this is a philosophical attitude i can have either one or the other i can go and say no my model uh but that uses all these tools all these mathematical and conceptual tools um is actually the how the brain is working and generating that data it's not only on my model it's actually how it is or i can say that no this is actually my tools that we as a scientific community have built to make sense of the data that we actually have access to and try to generate the the the activity that we don't have access to so we need to infer it right so just one comment on that particular distinction and then another thing that i wanted to um get to is that uh marco was saying oh i can't i can't see how we can how we don't have models how to get from one state to the other and that kind of thing i just want to say that in that regard that we don't have models we do have representations on the agent level i mean we think we use abductive reason we have we we use logics we have concepts we do make representations we represent things we can we can do that on the agent level so scientists do it we in our daily lives do that um and that is totally fine no one is disputing that the dispute is precisely on whether the underlying mechanisms of what allows for us to do that such as the body in the environment with the nervous system actually themselves also have that capacity so that's the question about models models on the agent level yes on the underlying mechanisms of what allows us to model that's a different story that i'm not convinced about um and then about how do we get like from one state to the other i think that we choose we think we use again uh reason abductive reason uh we have representation we also use all of

these intellectual machinery that we can and have because we are enculturated systems well and um and living beings read language and we choose how we get from one state to the next and then that can be all mapped out with you know dynamical causal models if we want to so i just wanted to finish with that thanks so much thanks everybody for this really fun discussion i'm glad that we opened so many great threads in point one i'm gonna provide a technical wish anyone can drop off they want to provide a technical wish and then a concluding thought so the one technical one is it'd be fun to have an automated for philosophical classification of statements even if only within the domain specific fep so a statement like the brain has to maximize the a matrix is a really if the brain is it is a policy selector or something like that is a realist take even if the person has some other perspective but in that sentence grammatically versus we use this version of spm to model this it's like it's an sentence so classifying sentences and then even just like kind of like a time series like shading our live stream by the way that people are speaking with an underlying generative model and then are they speaking from a position of implicitly like are people speaking from a position of um you know pronouns in one way or in some other way these are things that are like emitted states that could be inferred from linguistic patterns and then the concluding um thought was scott david talk about paradox he's talked about it before steven brought up that humans are like a tool culture everybody not just digital culture humans are like a handiwork tool people and so thomas said what do we lose with instrumentalism and so that's really just to conclude by saying that's what's on the table is do we just cut our ties with reality do we fight for realism and whatever that is lost along the way there or do we cut our ties with reality and go fully instrumentalist where does that leave us adrift if we have a really good ship where do we steer it because now we're going to be in more direct contact with our preferences with our governance structures because we're going to be really clear that science isn't like advancing some noble truth agenda per se it's something a lot more cns thank you it's something that's more on the um utilitarians the technological it's an development if the fep is within the realm of philosophy and engineering but not academia for example it's not too foreign of a world that could exist which would be different for an idea than other kinds of ideas and then it's the heart of the matter and is the heart of the matter how things are or is it how things are related to each other so that's which story we're in at that top level is the heart of the matter instrumentalism sidestepping the question you're losing everything because the heart of the matter is how things essentially are well instrumentalism is only sidestepping that target of what the heart of things are if you think that's essence but if you think that the heart of how

things are is relational then maybe you can be a realist about instrumentalism you say it's really i'm really making this scientific model it's really the code that i put on the this is really the conversation we're happy happening and all these sort of things like that so just it was really an awesome discussion much appreciated all of you active infra lobbies and uh we hope to see other people participate in the future if you're listening this far you probably could come on one of these conversations hopefully so thanks for participating everyone and the forum is in your calendar event for those who participated live everyone else just stay in touch but see you next time for 15.2